

SHAWINIGAN, QC, Canada

WMO: 713700

Lat: 46.564N

Lon: 72.733W

Elev: 110

StdP: 100.01

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-22.9	-31.1	0.2	-24.3	-28.6	0.3	-22.2	8.6	-8.7	7.7	-8.2	2.1	320	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.5	21.2	27.7	20.0	26.1	19.2	22.8	27.1	21.6	25.6	20.7	24.5	4.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.3	25.3	20.3	15.2	23.9	19.2	14.2	22.9	68.2	27.1	63.7	25.6	60.2	24.5	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.4	5.7	-28.7	32.5	2.6	2.0	-30.6	33.9	-32.2	35.1	-33.6	36.2	-35.5	37.7		
			-28.9	25.0	2.5	1.4	-30.7	26.0	-32.2	26.8	-33.6	27.6	-35.4	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-11.5	-9.5	-3.5	4.4	12.2	17.1	19.9	18.9	14.5	7.4	0.5	-7.3
	DBStd	11.93	6.91	6.11	5.88	4.42	4.14	3.68	2.89	2.91	3.97	4.09	4.72	6.08
	HDD10.0	2757	666	546	417	175	22	1	0	0	6	102	286	537
	HDD18.3	4902	924	779	676	419	196	67	16	29	126	339	536	795
	CDD10.0	1054	0	0	0	6	90	213	307	274	142	22	1	0
	CDD18.3	157	0	0	0	0	6	29	64	46	12	0	0	0
	CDH23.3	1212	0	0	0	4	87	257	480	296	87	1	0	0
	CDH26.7	277	0	0	0	0	21	67	117	52	19	0	0	0
Wind	WSAvg	2.7	2.6	2.8	3.0	3.1	2.9	2.6	2.4	2.4	2.4	2.7	2.7	2.7
Precipitation	PrecAvg	1111	76	71	74	85	95	100	98	103	100	100	97	111
	PrecMax	1416	123	209	153	243	267	251	185	223	180	276	184	191
	PrecMin	885	22	4	13	13	29	47	10	31	11	21	36	28
	PrecStd	142	26	49	31	47	51	41	39	46	36	52	37	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	4.9	12.3	21.6	29.4	31.2	32.1	30.2	29.4	22.1	14.2	5.9
		MCWB	3.8	2.5	6.3	12.9	18.6	21.4	22.3	21.5	21.7	16.8	11.5	4.9
	2%	DB	2.7	2.8	7.9	17.5	25.9	28.6	29.6	28.0	25.6	18.5	11.1	3.7
		MCWB	1.9	1.0	3.5	9.8	16.5	20.1	21.7	20.5	19.9	13.7	9.0	2.7
	5%	DB	0.9	1.3	5.8	14.4	22.7	26.2	27.7	26.4	23.5	15.7	8.9	2.3
		MCWB	0.2	-0.1	2.6	8.1	14.8	18.6	20.4	19.7	18.3	12.1	7.1	1.4
	10%	DB	-1.3	-0.2	4.1	11.5	20.1	24.2	26.0	24.9	21.3	13.6	6.7	0.8
		MCWB	-2.2	-1.3	1.5	6.4	13.4	17.3	19.4	19.1	17.1	11.0	5.1	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	3.5	7.7	15.3	20.6	23.5	24.7	23.5	22.7	17.8	12.4	5.3
		MCDB	4.4	4.3	10.1	21.5	26.2	28.8	29.8	27.7	26.6	21.0	13.3	5.7
	2%	WB	2.1	1.3	4.6	11.3	18.4	21.6	23.1	22.1	20.8	14.8	9.6	3.0
		MCDB	2.4	2.1	7.0	15.1	23.3	26.8	27.3	25.8	24.5	17.2	10.6	3.5
	5%	WB	0.3	0.2	3.1	8.8	16.5	20.1	21.9	21.1	19.4	12.9	7.3	1.5
		MCDB	1.0	1.1	5.0	13.1	20.1	24.1	25.6	24.7	22.2	14.8	8.7	2.1
	10%	WB	-2.2	-1.4	1.9	7.1	14.3	18.7	20.8	20.1	18.0	11.3	5.4	0.2
		MCDB	-1.3	-0.1	3.7	10.7	18.8	22.5	24.5	23.6	20.9	13.2	6.5	0.8
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.6	9.1	9.3	10.9	10.4	10.2	10.0	10.0	7.9	6.6	7.1
		MCDBR	8.5	8.6	10.0	14.9	15.4	13.7	12.3	12.0	12.3	11.5	9.6	6.8
		MCWBR	8.3	7.4	7.1	9.1	8.0	7.0	6.1	6.2	7.2	7.5	8.1	6.7
	5% WB	MCDBR	8.3	8.1	8.8	12.9	12.9	11.7	10.8	10.3	10.8	10.0	8.9	6.8
		MCWBR	8.2	7.5	6.8	8.9	8.1	7.2	6.3	6.0	6.9	7.4	8.1	6.9
Clear-Sky Solar Irradiance	taub	0.258	0.278	0.315	0.345	0.366	0.405	0.407	0.398	0.368	0.332	0.300	0.270	
	taud	2.318	2.248	2.256	2.382	2.401	2.284	2.293	2.325	2.399	2.506	2.504	2.356	
	Ebn at Noon	845	895	909	912	899	861	854	847	844	830	795	785	
	Edh at Noon	74	98	115	112	114	129	126	118	100	77	64	64	
All-Sky Solar Radiation	RadAvg	1.41	2.30	3.61	4.52	5.13	5.60	5.51	4.82	3.76	2.21	1.31	0.95	
	RadStd	0.20	0.27	0.33	0.44	0.54	0.41	0.51	0.23	0.30	0.17	0.17	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page